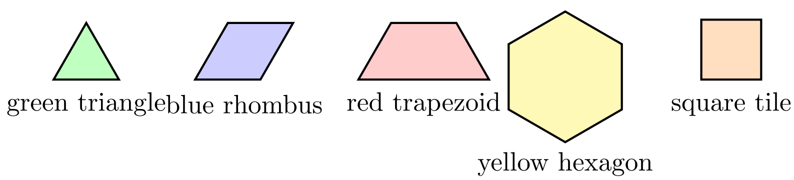
Putting Small Shapes Together to Make Bigger Shapes

Count the pieces that cover a bigger shape, name the new shape two pieces make, and compare two ways of covering the same shape

Directions:

- Use real pattern blocks if you have them. The picture below shows what each piece is called. **No numbers are shown on it** — the numbers you need are written inside each problem.
- When you *cover* a shape, the pieces must fit inside with no gaps and no pieces on top of each other.
- Some problems ask *how many* pieces. Some ask you to *draw and name* the new shape. Some ask you to compare two ways of covering. Read carefully and choose what to do.
- You may draw the lines between the pieces to help you count.



The name of every piece used on this page. No numbers are shown here: use the numbers written in each problem.

1. Rosa covers a rectangle with square tiles. She makes 2 rows, and each row has 4 tiles. How many square tiles cover the whole rectangle?

Answer: ________________

2. A yellow hexagon is covered exactly by 3 blue rhombuses. Each blue rhombus is covered by 2 green triangles. How many green triangles cover the whole hexagon?

Answer: ________________

3. Two identical squares are pushed together so that one whole side of each one touches. Draw the new shape they make. Then write the name of the new shape.

Answer: ________________

4. Ben covers a hexagon with 6 green triangles. Mia covers the very same hexagon with 3 blue rhombuses. Who used more pieces?

Write the two counts with $>$ or $<$ between them.

Answer: ________________

5. Every red trapezoid is covered by 3 green triangles. Nina builds 4 red trapezoids. How many green triangles does she use in all?

Answer: ________________

6. Two identical right triangles are put together so that their longest sides touch exactly. Draw the shape they make and write its name.

Then draw a *different* shape you can make out of the same two triangles.

Answer: ________________

7. The same big rectangle can be covered by 8 square tiles, or it can be covered by 16 triangle tiles. Which covering uses fewer pieces?

Write the two counts with $>$ or $<$ between them.

Answer: ________________

8. Ari covers one hexagon with 2 red trapezoids. Each red trapezoid is covered by 3 green triangles. Ari builds 2 hexagons the very same way. How many green triangles does he use in all?

Answer: ________________

9. Use four identical green triangles to build one larger triangle. Draw your build and show the lines between the pieces.

Then explain how your pieces show that the new shape really is a triangle.

Answer: ________________

10. **Challenge.** One hexagon can be covered three different ways: with 2 red trapezoids, with 3 blue rhombuses, or with 6 green triangles.

Put the three coverings in order from the *fewest* pieces to the *most* pieces. Write the three counts in that order.

Answer: ____________________